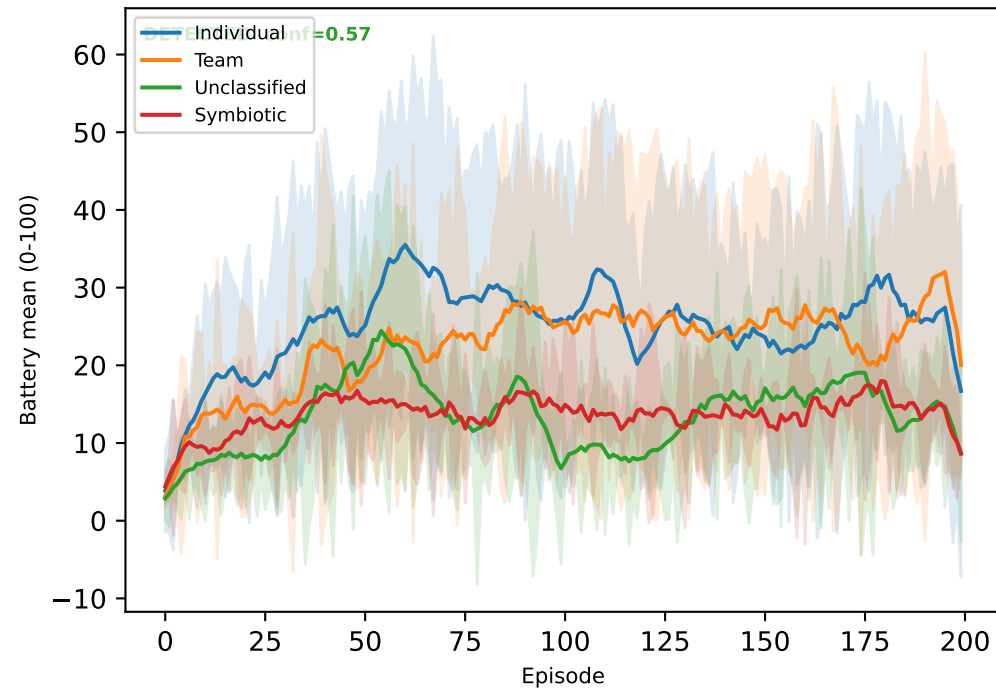
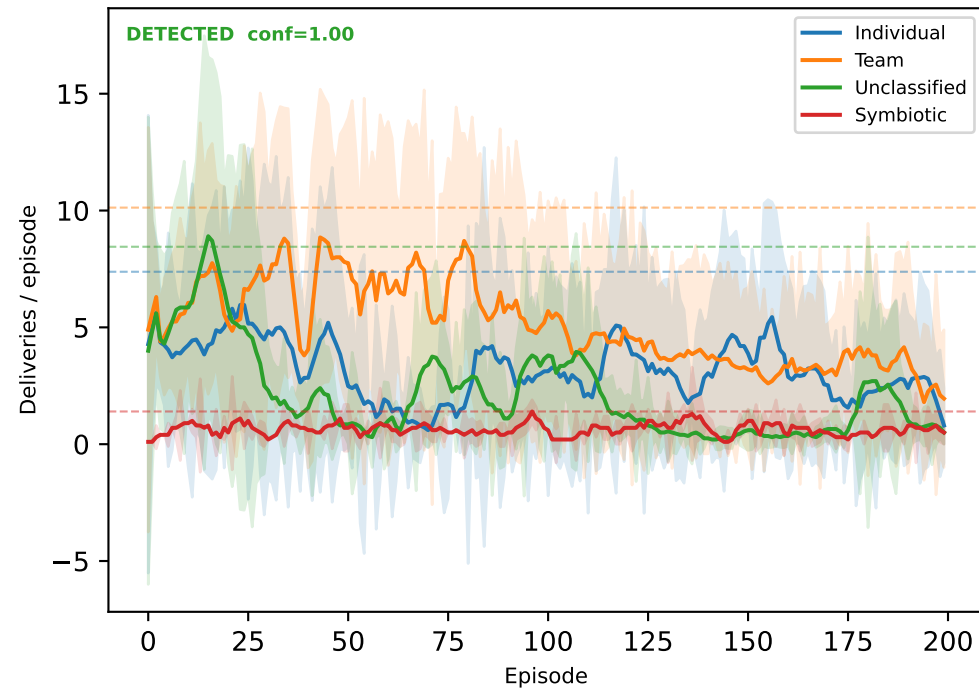


Emergent Behavior Analysis — TARWARE MARL Outstanding Runs

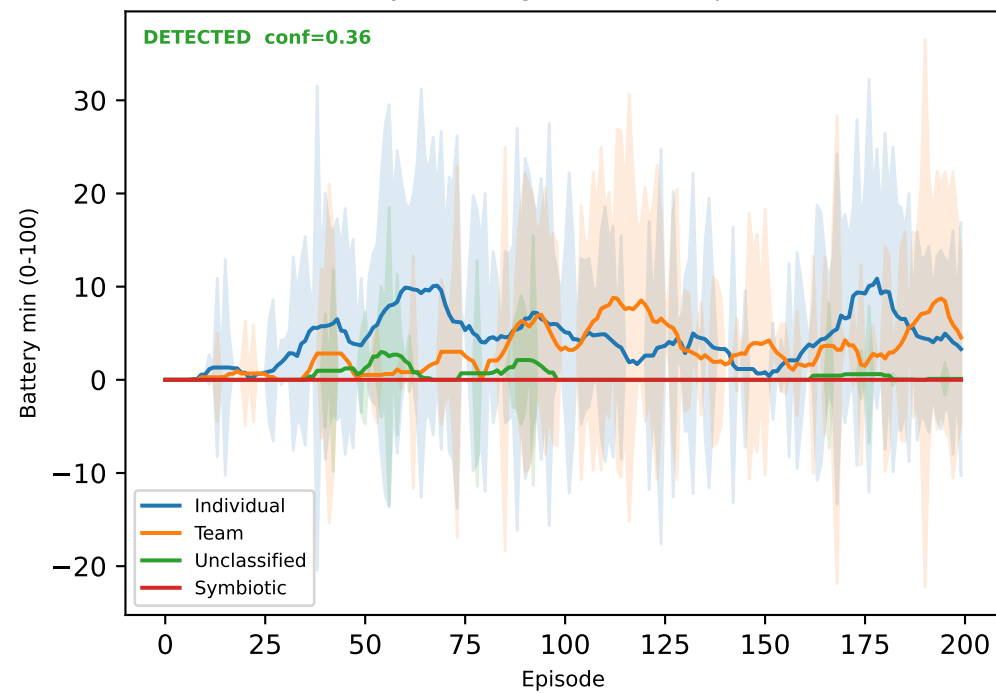
1. Energy-Aware Cooperation
(battery mean over training)



2. Spontaneous Task Redistribution
(deliveries per episode)



3. Reduced Conflict
(battery min — higher = fewer depletions)



4. Role Differentiation
(TSI · RSI · AGV-Picker entropy asymmetry)

